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The Double-Strand-Break Repair Model for Recombination

Review

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Summary

Gene conversion is the nonreciprocal transfer of information from one DNA duplex to another; in meiosis, it is frequently associated with crossing-over. We review the genetic properties of meiotic recombination and previous models of conversion and crossing-over. In these models, recombination is initiated by single-strand nicks, and heteroduplex DNA is generated. Gene conversion is explained by the repair of mismatches present in heteroduplex DNA. We propose a new mechanism for meiotic recombination, in which events are initiated by double-strand breaks that are enlarged to double-strand gaps. Gene conversion can then occur by the repair of a double-strand gap, and postmeiotic segregation can result from heteroduplex DNA formed at the boundaries of the gap-repair region. The repair of double-strand gaps is an efficient process in yeast, and is known to be associated with crossing-over. The genetic implications of the double-strand-break repair model are explored.

Introduction

Genetic recombination in eucaryotes has been studied in most detail in lower order fungi, because these organisms allow the recovery and analysis of all of the products of individual meiotic recombination events. These studies, reviewed below, have led to the proposal of several models for the mechanism of gene conversion and crossing-over. The most plausible of these models have involved initiation by a single-strand nick, followed by the generation of heteroduplex DNA. We propose a new model for meiotic recombination, based upon initiation by double-strand breaks and the repair of double-strand gaps. We first review the extensive genetic analysis of meiotic recombination in relation to earlier models, and then discuss the double-strand-break repair model and its genetic implications. Due to consideration of space, our review does not contain a thorough point-by-point referral to the original literature. More fully documented accounts have been given by Stahl (1979) and Whitehouse (1982). We would, however, like to acknowledge the contributions of those investigators whose extensive analysis of segregation in the fungi has provided the facts upon which this discussion of our model is based.

Conversion, Postmeiotic Segregation, and Crossing-Over

Upon the completion of premeiotic DNA replication, a diploid cell contains four DNA duplexes. The examination of eight-spored asci (Ascobolus, Neurospora, Sordoria) or of sectored-spore clones (Saccharomyces, Schizosaccharomyces) allows one to determine the genetic content of each of the eight single strands present at the beginning of meiosis. We will discuss all recombination in terms of these eight meiotic products. Any heterozygous marker will normally segregate $4^+:4^-$. The examination by tetrad analysis of all of the products of individual meiotic recombination events shows that recombinants between distant markers are produced in pairs (Figure 1a), with all markers showing 4:4 segregation. Such recombination events are called reciprocal exchanges or crossovers.

Occasionally a heterozygous marker will not segregate 4:4, but will show some aberrant pattern of segregation. The most common type of aberrant segregation in yeast is 6:2 (by which we mean $6^+:2^-$ or $2^+:6^-$) segregation, referred to here as gene conversion. Gene conversion reflects the nonreciprocal transfer of information from one duplex to another duplex (Figure 1b). The other two classes of aberrant segregation involve postmeiotic segregation. Postmeiotic segregation occurs when the two strands of a duplex carry different genetic information (heteroduplex DNA), so that a spore is produced that divides to make two genetically different daughter cells. Postmeiotic segregation results in sectored-spore colonies in yeast, and in nonidentical sister spores in the eight-spored fungi. If one sectored-spore clone or one pair of nonidentical sister spores is observed in an ascus, the segregation is referred to as a 5:3 segregation. If two sectored-spore clones or two pairs of nonidentical sister spores are found, the segregation is referred to as an aberrant 4:4 segregation. Both 5:3 and aberrant 4:4 segregations imply the existence of heteroduplex DNA. Heteroduplex DNA that is present on only one chromatid is called asymmetric heteroduplex DNA; heteroduplex DNA that covers the same region of two chromatids is called symmetric heteroduplex DNA. We will use recombination as a general term that includes all exchange of information between chromatids: crossing-over, gene conversion, and postmeiotic segregation.

Aberrant segregations do not arise from the interaction of two chromatids at a single point, but are due to an exchange of information over a short length of DNA. Thus, if a particular marker shows conversion, a nearby marker (i.e., in the same gene) will frequently be converted along with it (coconversion), but markers in an adjacent gene will almost always show normal 4:4 behavior.

Extensive tetrad analysis in yeast and other lower order fungi has shown that aberrant segregation of any kind is

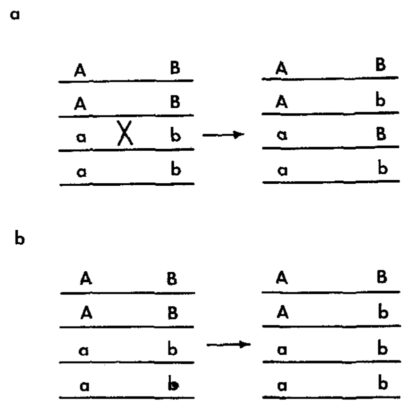

Figure 1. Crossing-Over and Conversion

(a) Crossing-over results in the production of complementary pairs of recombinant chromatids, with both markers segregating 4:4.

(b) Gene conversion results in the 6:2 segregation of one marker, and is therefore a nonreciprocal transfer of information from one chromatid to another.

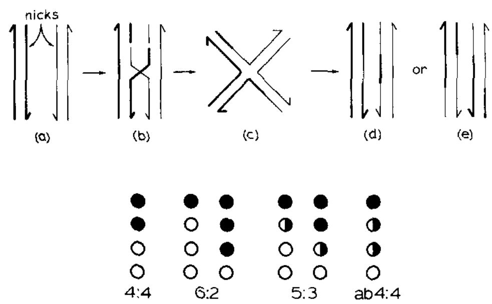

Figure 2. The Holliday Model

(Top) (a) Strands of the same polarity are nicked at homologous sites. (b) The nicked strands are exchanged on one side of the nick, forming symmetric heteroduplex DNA and a Holliday junction. (c) The Holliday junction can isomerize through a symmetrical intermediate, allowing resolution to occur by cutting either the originally crossed strands (d) or noncrossed strands (e).

(Bottom) The genetic consequences of the involvement of a single heterozygous site in symmetrical heteroduplex DNA generated by the Holliday model. When a marked site (solid circles and half-circles) falls within the region of heteroduplex DNA, mismatches are created and aberrant segregation can occur. If the mismatches on both DNA duplexes are corrected (repaired) in the same direction (e.g., mutant to wild-type) 6:2 segregation results; if the mismatch on one duplex is corrected, 5:3 segregation occurs. If neither mismatch is corrected, the outcome is aberrant 4:4 (ab4:4) segregation. In all cases, resolution of the Holliday junction may occur either by cutting the inner, crossed strands, in which case no crossover occurs, or by cutting the outer, noncrossed strands, in which case a crossover does occur. Thus the association of aberrant segregation and crossing-over is a consequence of the resolution of a Holliday junction at one end of a region of symmetric heteroduplex DNA.

associated with a high probability of a nearby reciprocal exchange. When a site shows conversion or postmeiotic segregation, properly segregating sites on either side show crossing-over at a frequency as high as 50%, even when these markers are closely linked. This association between aberrant segregation and crossing-over suggests that chromatids interact over limited regions such that both aberrant segregation and a high frequency of crossing-over of flanking markers can result. A knowledge of the exact properties of aberrant segregation (i.e., gene conversion and postmeiotic segregation) is therefore crucial to an understanding of the mechanism of recombination.

Aberrant segregation does not occur at a uniform frequency along the chromosome. The frequency of aberrant segregation for markers within a gene generally follows a gradient from a high value at one end of the gene to a low value at the other end. The simplest explanation for this phenomenon, termed polarity, is that interactions between chromatids start at special sites and then extend away from such sites over distances of one or a few genes.

The Holliday model (Holliday, 1964, 1968) was the first widely accepted molecular explanation of the relationship between aberrant segregation and crossing-over. The physical events of this model, including initiation by nicking, strand exchange to generate symmetric heteroduplex DNA, and Holliday junction formation and resolution, are described in Figure 2a. The genetic data explained by the physical model include the existence of the various types of aberrant segregation and the association of aberrant segregation and crossing-over (Figure 2b). Gene conversion arises from the repair of mismatches on both chromatids in a region of symmetric heteroduplex DNA. However, after the proposal of the Holliday model, the results of several experiments demonstrated that not all heteroduplex DNA was symmetric (see Figure 3 for details).

The Meselson–Radding Model

Meselson and Radding (1975) proposed a molecular model for the formation of asymmetric heteroduplex DNA (Figure 4). This model accounts for the data from Ascobolus by having asymmetric heteroduplex DNA, symmetric heteroduplex DNA, and crossing-over all follow from one initiation event. In yeast, asymmetric heteroduplex DNA is assumed to make a much greater contribution to the observed aberrant segregation than does symmetric heteroduplex DNA. In both cases, 6:2 segregation (gene conversion) results from the mismatch repair of heteroduplex DNA.

Many of the predictions of the Meselson–Radding model have been verified by Rossignol and his colleagues with the *b2* locus of Ascobolus immersus (Rossignol, 1969; Leblon and Rossignol, 1973; Rossignol et al., 1978). Within any one class of mutations, the overall frequency of aberrant segregation was seen to decrease from one end of the gene to the other, suggesting the presence of a site for the initiation of recombination near the high-conversion end of the gene. Moreover, base substitution and some frameshift mutations near the presumed site of initiation gave mainly 5:3 segregation and very few aberrant 4:4 segregants. Mutations towards the other end of the gene gave increasingly higher proportions of aberrant 4:4 segregation relative to 5:3 segregation. This result is consistent with the presence of asymmetric heteroduplex DNA close to the site of initiation of recombination, and a transition to symmetric heteroduplex DNA at some variable distance from the initiation site.

This interpretation of the data is strengthened by an analysis of the effects of an apparent deletion mutation

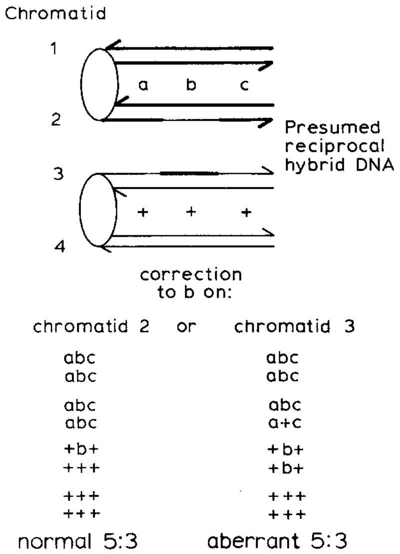

Figure 3. The Two Kinds of Noncrossover Asci Predicted by the Holliday Model

The flanking markers (a and c) are required to distinguish crossover from noncrossover asci. Correction to b on chromatid 2 results in normal 5:3 segregation; correction to b on chromatid 3 leads to aberrant 5:3 segregation. In Ascobolus, Stadler and Towe (1971) examined 5:3 asci that were nonrecombinant for markers flanking the aberrantly segregating site. They found that almost all of these asci were normal 5:3, and very few were aberrant 5:3. Stadler and Towe concluded that heteroduplex DNA is frequently formed on only one chromatid, at least when the flanking markers remain in the parental configuration. In yeast (Saccharomyces cerevisiae), several observations implied the presence of asymmetric, and the absence of symmetric, heteroduplex DNA (Fogel et al., 1978). First, aberrant 4:4 segregation is not observed, even for the few markers with a high frequency of postmeiotic segregation. Second, no excess of two-chromatid double crossovers within a short interval is seen. The Holliday model predicts such an excess because of mismatch correction within symmetric heteroduplex DNA. If the originally wild-type chromatid is corrected to mutant, and the originally mutant chromatid is corrected to wild-type, and the Holliday junction is resolved without crossing-over, the result is an apparent two-chromatid double crossover. Third, as in Ascobolus, significant numbers of aberrant 5:3 asci are not found.

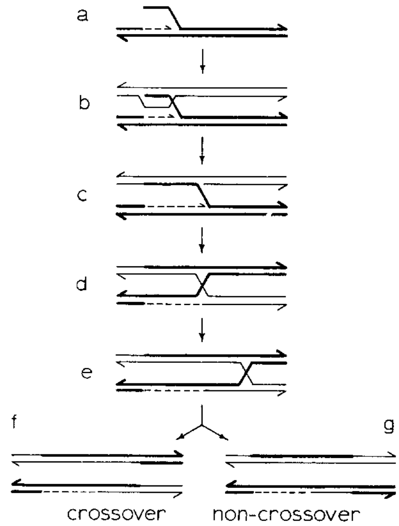

Figure 4. The Meselson–Radding Model

Recombination is initiated (a) by a single-strand nick on one of the two interacting duplexes. The 3′ end of the nicked strand acts as a primer for DNA synthesis, which displaces the strand ahead of it. The displaced single strand invades the other duplex at a homologous site, displacing a D loop (b) and forming a small region of asymmetric heteroduplex DNA. The single-stranded D loop is degraded, and the invading strand is ligated in place. The limited region of asymmetric heteroduplex DNA is expanded (c) by concerted DNA synthesis on the first (donor) duplex, and by exonucleolytic degradation on the second (recipient) duplex. After the enzymatically driven production of asymmetric heteroduplex DNA stops, either branch migration or isomerization can bring the 5′ and 3′ single-stranded ends into apposition so that they can be ligated. The resulting Holliday junction (d) can move along the duplex by the process of branch migration (rotary diffusion; Meselson, 1972), generating symmetric heteroduplex DNA (e). Resolution can yield either the crossover (f) or the noncrossover (g) configuration.

(G234) within the *b2* gene (Hamza et al., 1981). This mutation has no spore-color phenotype, and when homozygous, has no effect on patterns of aberrant segregation. However, when heterozygous, the mutation specifically causes a decrease in aberrant 4:4 segregation for base-substitution mutations that lie on the low-conversion side of the gene. Mutations on the high-conversion side are unaffected, as is 5:3 segregation of mutations on the low-conversion side of the gene. The simplest explanation of this result is that the heterology blocks the migration of a Holliday junction, but does not affect the enzyme-catalyzed formation of asymmetric heteroduplex DNA.

The Meselson–Radding model can explain all of the Ascobolus and yeast data described above. However, additional assumptions or modifications are required to explain the results of some more recent experiments (Fogel et al., 1978). These experiments include observations on the location of crossover events relative to conversion events and initiation sites (Figure 5), the parity of gene conversion for all classes of mutations in yeast, the apparent strand specificity of mismatch repair (Figure 6), and the fact that the strand upon which initiation occurs appears to be the recipient of genetic information (for review see Stahl, 1979).

These observations place specific constraints on several processes invoked by the Meselson–Radding model, e.g., branch migration, mismatch correction, and initiation. While none of these experiments in any sense disproves the Meselson–Radding model, they do make it worthwhile to consider alternative models that may provide a simpler explanation of the data.

The Double-Strand-Break Repair Model

The transformation of yeast with Escherichia coli plasmids containing fragments of yeast DNA provides a useful system for the study of recombination because one of the recombining molecules can be manipulated in vitro, and in some cases all of the products of a recombination event can be recovered. We will review the results of recent studies on plasmid–chromosome recombination in yeast (Orr-Weaver et al., 1981, 1983; Orr-Weaver and Szostak, 1983). These experiments establish the recombinogenic nature of double-strand breaks, and demonstrate the occurrence of gene conversion by double-strand-gap repair. We present a molecular model for the repair of double-

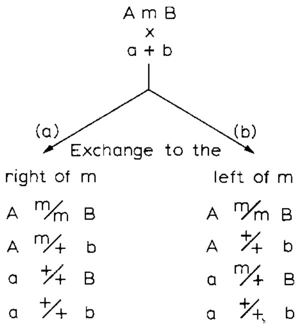

Figure 5. Crossover Position

In yeast, most mutations show predominantly the 6:2 type of aberrant segregation, but 5:3 segregation is seen and is also associated with a high frequency of crossing-over between flanking markers (a and b). Crossovers associated with postmeiotic segregation could occur on either the high-conversion side of the gene (towards the initiation site), or on the low-conversion side of the gene (away from the initiation site). The Meselson–Radding model predicts that the crossovers will occur on the low-conversion side of the gene; the two possibilities can be distinguished in crosses with flanking markers. Fogel et al. (1978) examined one site within the *arg4* locus, and found 44 crossovers on the high-conversion side and 20 crossovers on the low-conversion side of the aberrantly segregating site. This result can be explained within the context of the Meselson–Radding model by assuming that the Holliday junction can branch-migrate in only one direction. Branch migration away from the initiation site must not occur, since this would generate symmetric heteroduplex DNA (and hence aberrant 4:4 asci, which are not observed in yeast). However, branch migration back towards the initiation site could explain the observed crossover positions. Alternatively, initiation could involve an enzyme that acts like a type I restriction enzyme, i.e., an enzyme that would bind to a specific site, but would then slide along the DNA for a variable distance before nicking and initiating recombination. If asymmetric heteroduplex DNA could form either to the left or to the right, the final crossover could appear to lie on either side of a particular stretch of heteroduplex DNA.

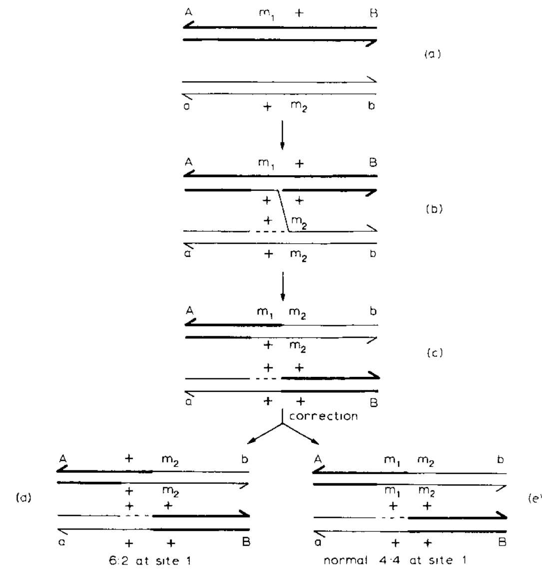

Figure 6. Close Marker Recombination

If mismatches in yeast have an equal probability of being corrected in either direction, the Meselson–Radding model predicts equal numbers of conversion and crossover asci for close markers. (a) The close markers m_1 and m_2 are flanked by the distant markers A and B. (b) If asymmetric heteroduplex DNA covers one of the close markers, and resolution occurs between the close markers (c) to yield the crossover configuration for the flanking markers, mismatch repair can lead to conversion (d) or crossing-over (e) for the close markers. Experiments by Savage and Hastings (1981) with the *his1* locus and by Fogel et al. (1978) with *arg4* suggest, within the context of the Meselson–Radding model, that mismatch correction must use only the invading strand as a template for repair, and that parity results from equal frequencies of initiation on either chromatid. Both groups of investigators performed crosses involving two markers within a gene, and two flanking markers. Asci were identified in which the intragenic marker proximal to the presumed initiation site had converted (6:2 segregation), while the intragenic marker distal to the initiation site showed normal 4:4 segregation, and the flanking markers had recombined. However, if mismatch repair can occur in either direction (to wild-type or mutant), a second class of asci should be generated. This second class would appear to have resulted from a simple crossover between the intragenic markers, with all markers showing normal 4:4 segregation. Since this class is almost never observed, mismatch repair to restore the original genotype of a chromatid appears to be uncommon compared with mismatch repair in favor of the invading strand.

strand breaks and gaps, and we discuss two questions: Could meiotic recombination be initiated by double-strand breaks? and, Could meiotic gene conversion in yeast occur by a process of double-strand-gap repair? We show that our model is in fact able to account for the properties of meiotic recombination reviewed above. Finally, we discuss our model in terms of other classes of recombination events in yeast that either do, or may, involve double-strand breaks, e.g., *MAT* switching, spontaneous and induced mitotic recombination, and *spm*-induced Ty1 excision.

Plasmid–Chromosome Recombination

The integration of hybrid plasmids into the yeast genome occurs by homologous recombination and is the basis of the yeast transformation technique developed by Hinnen et al. (1978). Transformation with closed circular plasmid DNA results in the recovery of about 0.1–1 transformant per microgram of DNA per kilobase of homology per 10^8 cells. Transformation is limited by recombination, since more transformants can be recovered by increasing the available homology (Szostak and Wu, 1979), or by using plasmids capable of autonomous replication (Beggs, 1978; Stinchcomb et al., 1979).

We have shown that the introduction of a double-strand break (by restriction endonuclease digestion) into a region of yeast DNA on a transforming plasmid leads to a stimulation by as much as 3000-fold in the frequency of transformation (Orr-Weaver et al., 1981). Furthermore, the structures of the integrated plasmids, as determined by Southern blot restriction mapping, are identical when cut and circular plasmid DNA are used for transformation. The integration of a cut plasmid results in the repair of the initial double-strand break, since the final structure does not contain any double-strand breaks. A plasmid containing two regions of homology with the yeast genome was shown to integrate at either homologous yeast locus when circular molecules were used for transformation. When the plasmid was cut with a restriction enzyme in a given region of homology, all integration occurred at the genomic locus homologous with the region of the plasmid containing the double-strand break. This type of "targeting" experiment shows that DNA ends interact directly with homologous sequences.

When a double-strand gap was created within a region of yeast DNA on a hybrid plasmid, by cleavage at two restriction sites, we found that the double-strand gap was always repaired from genomic sequences during plasmid integration. Transformation with gapped plasmids never results in the appearance of a deletion corresponding to the original gap in the integrated structure (Orr-Weaver et al., 1981). In contrast, a deletion on a plasmid is readily integrated when circular plasmid DNA is used for transformation. The fidelity of the double-strand repair process is demonstrated by the efficient regeneration of a functional gene from a gapped gene during plasmid integration. Since the repair of a double-strand gap involves the transfer of genetic information from one DNA duplex to another, gap repair can results in conversion (Figure 7).

Gene conversion by double-strand-gap repair has several properties that make it similar to meiotic gene conversion. First, it is associated approximately 50% of the time with crossing-over. When a gap is made on a plasmid that is capable of autonomous replication, and transformants resulting from correct gap repair are selected, similar numbers of both integrated and nonintegrated plasmids are recovered (Orr-Weaver and Szostak, 1983). Integrated plasmids correspond to gap repair with crossing-over, and nonintegrated plasmids correspond to gap repair without

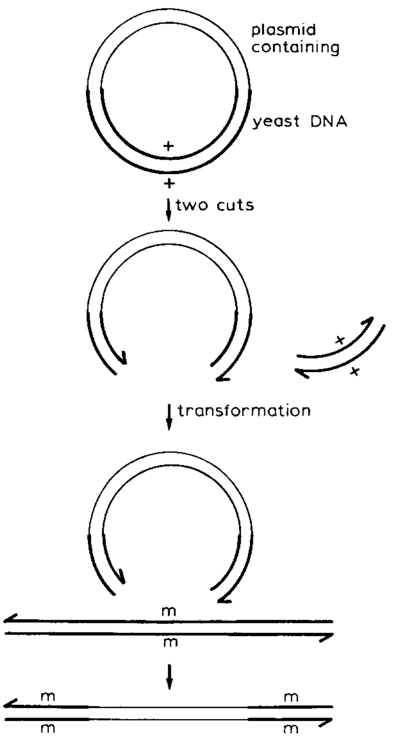

Figure 7. Double-Strand-Gap Repair in Yeast

An E. coli plasmid containing a segment of yeast DNA is cut with restriction enzymes to produce a double-strand gap within the yeast DNA. The gapped plasmid is efficiently integrated into the homologous chromosomal locus, and the gap is repaired from chromosomal information during integration.

crossing-over. Second, gap repair is associated with the formation of heteroduplex DNA adjacent to the DNA ends. When we transformed a *his3*$^-$ strain with a plasmid gapped in the *his3* gene, we recovered occasional transformants that had a functional chromosomal *HIS3*$^+$ gene, but an unstable (nonintegrated) plasmid. The simplest explanation for this result is the transfer of genetic information to the chromosome from plasmid sequences adjacent to the gap by the formation of heteroduplex DNA (Orr-Weaver and Szostak, 1983). The extent of this heteroduplex DNA and the fate of mismatches within it are currently being studied. Third, the processes of double-strand-break and -gap repair, like meiotic recombination, require the product of the *RAD52* gene (Resnick and Martin, 1976; Prakash et al., 1980; Game et al., 1980; Orr-Weaver et al., 1981). *RAD52*-defective mutants were originally identified because of their sensitivity to ionizing radiation. The *RAD52* gene product, while not characterized enzymologically, has been shown to be required for homology-dependent repair of double-strand breaks in yeast (Resnick and Martin, 1976). The many similarities between double-strand-break or -gap repair and meiotic recombination, and the fact that both processes require the *RAD52* gene product, have led us to consider the hypothesis that meiotic recombination is initiated by double-strand breaks.

A Molecular Model for Double-Strand-Break and -Gap Repair

The salient features of our model for double-strand-break and -gap repair are: the repair of a double-strand gap by two rounds of single-strand repair synthesis, the existence of regions of heteroduplex DNA flanking the region of gap repair, and the resolution of the repair event by the resolution of two Holliday junctions.

A specific version of a model with these features is illustrated in Figure 8. Recombination is initiated by the introduction of a double-strand break into the recipient chromatid by a double-strand endonuclease. The cut is enlarged to a gap, and 3′ single-stranded termini are produced by the action of exonucleases. One of the free 3′ ends then invades a homologous region of the donor duplex, displacing a small D loop. This D loop is enlarged by repair synthesis primed from the invading 3′ end. Eventually the D loop will contain sequences complementary to the 3′ end from the other side of the gap. The complementary single-stranded regions anneal, and a second round of repair synthesis is initiated from this 3′ end. In this way, the gap is repaired by two rounds of single-strand repair synthesis. Two regions of asymmetric heteroduplex have also been formed: one on the left by annealing of single strands, and one on the right by strand invasion. The lefthand region of asymmetric heteroduplex DNA may be extended enzymatically by concerted repair synthesis and exonucleolytic degradation. The righthand region of heteroduplex DNA may or may not be extended in the same way, depending on the ability of the repair synthesis reaction to drive branch migration of a junction with two crossed strands. After these reactions stop, branch migra-

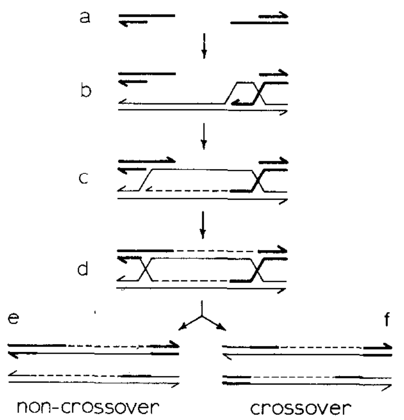

Figure 8. A Double-Strand-Break Repair Model

(a) A double-strand cut is made in one duplex, and a gap flanked by 3′ single strands is formed by the action of exonucleases. (b) One 3′ end invades a homologous duplex, displacing a D loop. (c) The D loop is enlarged by repair synthesis until the other 3′ end can anneal to complementary single-stranded sequences. (d) Repair synthesis from the second 3′ end completes the process of gap repair, and branch migration results in the formation of two Holliday junctions. Resolution of two junctions by cutting either inner or outer strands leads to two possible noncrossover (e) and two possible crossover (f) configurations. In the illustrated resolutions, the righthand junction was resolved by cutting the inner, crossed strands.

tion will result in the formation of two Holliday junctions. Branch migration of the Holliday junctions would generate symmetric heteroduplex DNA. Resolution of the two Holliday junctions in the same sense (i.e., cutting inner or outer strands in both cases) results in the noncrossover configuration of flanking markers; resolution in the opposite sense results in crossing-over.

The model described above accounts for the observed properties of meiotic recombination as follows. Gene conversion (6:2 segregation) can arise in two ways according to this model. If a marked site falls within a double-strand gap, it will be converted by a double-strand transfer of information, without the direct involvement of heteroduplex DNA. Alternatively, if the marked site falls within either of the flanking regions of asymmetric heteroduplex DNA, mismatch correction will lead to 6:2 segregation as in the Meselson–Radding model. Heteroduplex DNA in which a mismatch has not been repaired results in postmeiotic segregation. According to the model, the actual site of initiation is surrounded by a double-strand-gap repair region, which is flanked on both sides by asymmetric heteroduplex DNA, which is in turn flanked by (potential) symmetric heteroduplex DNA. This pattern can be varied to account for the patterns of aberrant segregation seen in both yeast and Ascobolus. For yeast, we assume that the size of the double-strand gap region is substantial, and that this region is flanked by relatively short regions of asymmetric heteroduplex DNA, sufficient to account for the observed levels of postmeiotic segregation. For Ascobolus, we need only assume that the double-strand gap is small relative to the amount of asymmetric and symmetric heteroduplex DNA. The size of the double-strand gap would depend on the activities of the exonucleases involved, and the rates of the strand invasion and annealing reactions. The relative amounts of asymmetric and symmetric heteroduplex DNA could be controlled by the length of the single-stranded DNA involved in strand invasion and annealing, by the action of repair synthesis in the extension of symmetric heteroduplex DNA, by the rates of branch migration, by isomerization, or by Holliday junction resolution.

Our model and the Meselson–Radding model explain several aspects of meiotic recombination in different ways. The Meselson–Radding model postulates that gene conversion (6:2 segregation) results from mismatch repair within regions of heteroduplex DNA. If this is so, the properties of the mismatch repair systems of yeast and Ascobolus must be quite different. In Ascobolus, insertion, deletion, and most frameshift mutations appear to be corrected efficiently, and base-substitution mutations appear to be corrected inefficiently. In yeast, all classes of mutations appear to be corrected efficiently, since all yield a large excess of 6:2 asci over 5:3 asci (Fogel et al., 1978). Furthermore, in Ascobolus, frameshift mutations show strong disparity, i.e., they are corrected preferentially to either wild-type or mutant, depending on the particular mutation (Rossignol and Paquette, 1979). However, all mutations in yeast show parity (Hurst et al., 1972; Fogel et al., 1978); i.e., gene conversion favors neither the wild-type nor the mutant allele. Gene conversion of large deletions in both Ascobolus (J. L. Rossignol, personal communication) and yeast shows parity (Fink and Styles, 1974; Lawrence et al., 1975). Radding (1978) has proposed a model for the conversion of deletions and insertions by a mechanism that involves the formation of heteroduplex DNA that includes a large heterology. If repair synthesis of the single-stranded loop always occurs, resolution of the resulting Holliday junction would lead, with equal probability, to conversion to wild-type or mutant.

If most gene conversion in yeast occurs by double-strand-gap repair, then parity in conversion simply reflects an equal probability of initiation on either chromatid. Since heteroduplex DNA and mismatch repair are not involved, no constraint on mismatch repair (in favor of the invading strand) is required, as it is for the Meselson–Radding model (Figure 6), and no excess of apparent two-strand double crossovers can arise (see Figure 3). Double-strand-gap repair provides a simple explanation for parity in the conversion of large deletions and insertions. The generation, on the wild-type chromatid, of a gap that covers the position of an insertion on the mutant chromatid will lead to conversion to the insertion after repair, since the only template available is the insertion chromatid. On the other hand, the formation, on the insertion chromatid, of a gap that overlaps the site of the insertion must lead to conversion to wild-type. This is because the insertion side of the gap cannot interact with the other chromatid until the insertion has been removed by continued exonucleolytic enlargment of the gap, so that sequences with homology to the other chromatid are exposed. The same reasoning applies to deletions.

According to the double-strand-gap repair model, initiation occurs on the chromatid that is the recipient of genetic information in a conversion event. In the Meselson–Radding model, initiation occurs by the introduction of a nick on the chromatid that is the donor of genetic information in a conversion event. Genetic evidence from three organisms suggests that initiation occurs on the recipient duplex. In Schizosaccharomyces pombe (Gutz, 1971) and Sordoria brevicollis (MacDonald and Whitehouse, 1979), unusual mutations have been found that display abnormally high levels of gene conversion. The simplest explanation of these mutations is that they fortuitously create new sites for the initiation of recombination. Both of these mutations show strong disparity, and are, when heterozygous, preferentially converted to wild-type; i.e., the chromosome on which initiation occurs is the recipient of genetic information. In Neurospora crassa (Catcheside and Angel, 1974) the genetic element cog^+ confers a high rate of conversion upon genes in its vicinity. In cog^+/cog^- heterozygotes, the cog^+ chromosome is usually the recipient in conversion events that are not accompanied by crossing-over. These results can be explained by modifying the Meselson–Radding model in either of two ways. Markham and Whitehouse (1982) have suggested that an initiation enzyme recognizes the new initiation site, and introduces a nick not at that site, but at a homologous site on the opposite chromatid. Alternatively, nicking could be followed by degradation to form a single-strand gap. Strand invasion would then lead to the formation of asymmetric heteroduplex DNA on the initiating duplex (Radding, 1983).

The observation that the frequency of aberrant segregation decreases from one end of a gene to the other (polarity) can be explained in either of two ways by both the double-strand-break repair model and the Meselson–Radding model. If initiation occurs at fixed sites, then polarity reflects the variable size of the regions of gap repair and heteroduplex DNA. Thus a site at an increasing distance from the initiation site has a decreasing probability of falling within a gap or within heteroduplex DNA. If initiation is a two-step process, in which an enzyme binds at a specific site, but moves along the DNA before cutting and initiating recombination, then polarity reflects a decreasing probability of initiation with increasing distance from the central binding site. The fixed-site model predicts a decreasing probability of 6:2, and an increasing proportion of 5:3, segregation as a function of distance from the initiation site. While the limited data available from the *arg4* locus (Fogel et al., 1978) do not support this view, data from many more alleles will be required to reach a definite conclusion. The two-step model would tend to obscure such a gradient.

The association of crossing-over with conversion is a natural consequence of the resolution of a double-strand-break repair event by the resolution of two Holliday junctions. This association has been experimentally verified in the plasmid–chromosome recombination system (Orr-Weaver and Szostak, 1983). Since the two Holliday junctions to be resolved flank the site of gene conversion, the crossover may appear genetically to lie on either side of the conversion site, as observed by Fogel et al. (1978).

Recent electron microscopic observations on DNA from meiotic yeast cells are consistent with the appearance of Holliday junctions in pairs. Bell and Byers (1982) isolated DNA from meiotic cells at the time of recombination, crosslinked the strands to prevent branch migration, cleaved the DNA with a restriction enzyme, and examined it in the electron microscope. Most chi forms (molecules joined at homologous sites) were joined not by a single Holliday junction, but by a fused region or by two junctions separated by 100–1000 bp. These molecules, if indeed they represent true recombination intermediates, are clearly unexpected by the Meselson–Radding model, but are explained simply by the double-strand-break repair model.

The version of the double-strand-break repair model illustrated in Figure 8 does not explain the absence of aberrant 5:3 segregation (see Figure 3) in yeast. Indeed, consideration of the fate of heterozygous sites lying within each of the two regions by asymmetric heteroduplex DNA (Figure 8) leads to the conclusion that, in the absence of mismatch repair, at least one of the two sites must result in an aberrant 5:3 segregation. In Figures 9 and 10, we present two models that avoid the formation of aberrant 5:3 asci.

Other Aspects of Meiotic Recombination

The *arg4-16* allele shows an unusually high level, for yeast, of postmeiotic segregation (about 30% of total aberrant segregation), even though it maps between two alleles that have low levels (about 5%) of postmeiotic segregation (Fogel et al., 1978). This is explained within the Meselson–Radding model by assuming that *arg4-16* is an unusual allele that is not efficiently repaired. The behavior of the *arg4-16* allele is most simply explained within the double-strand-break repair model by the assumption that the mutation creates a pause site for an exonuclease. If, during the enlargement of a double-strand gap by an exonuclease, a pause site is reached, sequences adjacent to that site will have a high probability of becoming involved in the formation of heteroduplex DNA and the generation of postmeiotic segregation. Pause sites have been observed in the action of several exonucleases, including Exo III (Shalloway et al., 1980), lambda exonuclease, Bal 31 and T4 polymerase exonuclease (Yang et al., 1983), and Exo VIII (J. Joseph, personal communication).

M. Williamson and S. Fogel (personal communication) have isolated mutations in a set of genes (*cor1*, *cor2*, *cor3*) that they propose are required for mismatch repair (see also Fogel et al., 1982). Mutations in these genes lead to a dramatic elevation in the level of postmeiotic segregation for many mutations. The interpretation of this phenotype in terms of the Meselson–Radding model is that mismatches in heteroduplex DNA, which would normally be corrected to give 6:2 segregations, are instead left unrepaired, so that 5:3 segregation occurs. This explanation

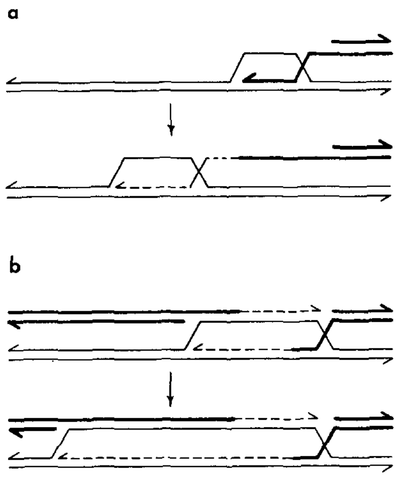

Figure 9. Position and Fate of Heteroduplex DNA

Initiation as shown in Figure 8 creates heteroduplex DNA on the righthand side of the gap by strand invasion, and on the lefthand side of the gap by annealing. To avoid the formation of aberrant 5:3 asci, all heteroduplex DNA must be confined to one chromatid, and one Holliday junction must always be resolved in a defined way. The righthand heteroduplex DNA may be removed by single-stranded branch migration as the first round of repair synthesis is in progress (a). Alternatively, the righthand region of heteroduplex DNA may be quite short, and the lefthand region may be extended enzymatically, so that essentially all observed postmeiotic segregation derives from the lefthand heteroduplex DNA (b).

There are two plausible ways in which the resolution of the Holliday junctions could be constrained so as to avoid aberrant 5:3 segregation. The models of initiation shown confine all heteroduplex DNA to the recipient chromatid. In this case, one Holliday junction must be resolved by cutting the inner (crossed) strands. Such a bias on resolution could derive from the nature of a pair of adjacent Holliday junctions. The two Holliday junctions constrain each other and prevent isomerization, since neither is free to rotate into the open configuration. Once one junction has been resolved by an enzyme that cuts only crossed strands, the second junction is free to isomerize, and can be resolved in either plane.

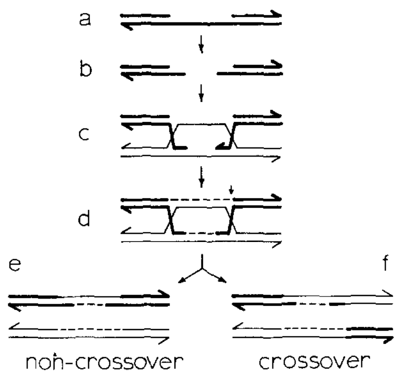

Figure 10. A Double-Strand-Break Repair Model in Which All Asymmetric Heteroduplex DNA Is Confined to One Chromatid

Initiation occurs by the formation of a single-strand gap (a), which is then processed into a double-strand gap by endonucleases and exonucleases (b). Both the 5′ and the 3′ single-stranded ends invade the homologous duplex (c), and the double-strand gap is repaired by two rounds of single-stranded repair synthesis (d). This initiation model confines all heteroduplex DNA to the donor chromatid; to avoid the generation of aberrant 5:3 asci, one Holliday junction must be resolved by cutting the outer (noncrossed) strands. After the completion of the second round of repair synthesis, one of the outer strands of the righthand Holliday junction is already nicked. This nick could bias the resolution of the Holliday junction towards cutting of the outer strands. Resolution of the lefthand junction can then result in a noncrossover (e) or crossover (f) configurations.

predicts that the level of gene conversion will fall in proportion to the stimulation of postmeiotic segregation. The published data show that in some cases the level of gene conversion falls, while in other cases it is unchanged, and in some cases conversion is increased at the same time that postmeiotic segregation increases. The recombination phenotype of the *cor* mutants can be explained, within the context of the double-strand-break repair model, by assuming that they affect the activity of the exonucleases that create the double-strand gaps. Increased 5′ to 3′ exonuclease activity would increase the size of the single-stranded regions available for the formation of heteroduplex DNA, without affecting the size of the double-strand gap. Alternatively, a defective single-strand-specific exonuclease could increase the size of the single-stranded tails at the expense of the double-strand gap region.

Other Recombination Events in Yeast

Mating-type switching in yeast (for review see Herskowitz and Oshima, 1981) is initiated by the enzymatic production of a specific double-strand break within the mating-type locus (Strathern et al., 1982). Mating-type switching is a process of gene conversion in which a block of DNA at the *MAT* locus is replaced by a block of DNA from one of the silent mating-type cassettes, *HMR* or *HML* (Klar et al., 1980; Haber et al., 1980). This conversion event is unusual because of its frequency (as high as once per cell cycle), and because it is almost always resolved without crossing-over. Resolution with crossing-over is lethal, since it generates either a large deletion (Hawthorne, 1963) or a circular chromosome (Strathern et al., 1979). The switching process also shows extreme disparity. It is always the *MAT* locus, and not a silent casette, that is the recipient of information. Efficient switching requires the product of the *HO* gene. It now appears that *HO* cells produce an endonuclease that makes a specific double-strand break in the *MAT* locus, but not in the silent cassettes (R. Kostrikan and F. Heffron, personal communication).

A simple double-strand-break repair model for mating-type switching is that the initial double-strand break is enlarged to a gap by degradation of the old *MAT* information, and that the gap is subsequently repaired using one of the silent cassettes as a template. The high frequency of switching reflects the efficiency of cutting within *MAT*, and the disparity reflects the fact that the *MAT* locus, and not the silent cassettes, is cut. Attempted switching in *rad52* cells is lethal (Malone and Esposito, 1980; Weiffenbach and Haber, 1981), presumably because all double-strand-break repair requires the *RAD52* gene product. The low frequency of crossing-over associated with *MAT* switching must mean that *MAT* switching is resolved differently than other double-strand-break repair processes. If two Holliday junctions are formed, as in our model, crossing-over would be suppressed if both junctions were always resolved in the same plane. This could be achieved if the resolution enzyme cut, for example, only the crossed strands, and if isomerization were suppressed. If there were insufficient time for isomerization to occur between the cutting of the first and second junctions, crossing-over would be less frequent. Alternatively, the double Holliday structure could be resolved by replication without crossing-over.

Both radiation-induced and spontaneous mitotic gene conversion requires the *RAD52* gene product. Resnick and Haber (unpublished data; Haber et al., 1982) have studied gamma-induced gene conversion at the *lys2* locus. They irradiated a haploid *lys2* strain, mated it to a haploid strain bearing a different *lys2*, allele, and selected LYS^+ diploids. They found that it was almost always the irradiated *lys2* allele that was lost in the diploid; i.e., the chromatid on which initiation took place was the recipient of genetic information. Furthermore, at least 15% of the conversion events were accompanied by crossing-over. Spontaneous mitotic recombination has been the subject of detailed genetic analyses (Esposito, 1978; Roman, 1980). Esposito examined conversion events that resulted in the formation of sectored colonies, and concluded that conversion resulted from mismatch repair in heteroduplex DNA formed in G1 of the cell cycle, and that Holliday-junction resolution occurred by replication. Roman examined unselected convertants and concluded that most events were both initiated and resolved in G1. Roman also noted that his data could be explained either by mismatch repair in heteroduplex DNA or by a two-strand transfer of information. Thus it is possible that at least some fraction of the spontaneous level of mitotic gene conversion actually reflects the repair of spontaneous double-strand breaks. It is interesting to note that the integration of circular plasmids (Orr-Weaver et al., 1981), the excision of integrated plasmids (Jackson and Fink, 1981), and spontaneous sister-chromatid exchange (Prakash and Taillon-Miller, 1980) do not require *RAD52*. There may be a second, different pathway of recombination active in mitotic cells that does not involve double-strand breaks and does not require the *RAD52* gene product.

Reversion of certain mutations caused by the insertion of the Ty1 transposon occurs most frequently by recombination between the directly repeated delta elements that flank the transposon, and is *RAD52*-independent. However, the enhanced delta–delta recombination observed in *spm* mutant strains does require *RAD52* (Roeder and Fink, 1983). This suggests a different mechanism for *spm*-mediated delta–delta recombination—perhaps a mechanism associated with the site-specific introduction of double-strand breaks.

The double-strand-break repair model is similar to certain models for transposition in that an event is initiated by a double-strand break at the recipient site, and the resulting DNA ends act as primers for the replication of donor sequences (Harshey and Bukhari, 1981). This similarity could reflect a common evolutionary origin for the processes of homologous recombination and transposition.

Conclusions

We have reviewed the experimental data on the genetic properties of meiotic recombination and double-strand-break repair. We have proposed a molecular model for double-strand-break repair, and we have shown that this model is consistent with the genetic properties of meiotic recombination. We therefore conclude that meiotic recombination may be initiated by double-strand breaks. The ability to repair double-strand breaks in a homology-dependent reaction is a very efficient process in yeast (Resnick and Martin, 1976). This repair is recombinogenic, and may have been adapted for use in meiotic recombination during the evolution of meiosis. Yeast mating-type switching is initiated by a specific double-strand break; here again, the double-strand-break repair pathway seems to have been co-opted evolutionarily for a task requiring recombination. The existence in Drosophila of mutants (*mei9*, *mei41*) that are similar to *rad52* in that they are x-ray-sensitive and defective in meiotic recombination (Baker et al., 1978) leads us to speculate that meiotic recombination in Drosophila, and perhaps other organisms, also involves double-strand-break repair. Since many of the genetic predictions of the double-strand-break repair model are similar to those of the Meselson–Radding model for recombination, we suggest that biochemical experiments will be necessary to determine the actual mechanism of initiation of meiotic recombination. The most critical experiment is a direct search for the appearance of double-strand breaks during the period of commitment to meiotic recombination. Another important experimental direction is the analysis of the *cor* mutants. Synthetic mismatches, such as have been made with phage λ (Wildenberg and Meselson, 1975), could be used to test the involvement of the *COR* gene products in mismatch repair. Eventually, enzymological analysis of gene products shown by genetic criteria to be involved in meiotic recombination should allow a detailed reconstruction of the initiation process.

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